

Dynamically vulcanized PP/EPDM blends with balanced stiffness and toughness via in-situ compatibilization of MAA and excess ZnO nanoparticles: Preparation, structure and properties

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ABSTRACT

It is a challenge to simultaneously improve the stiffness and toughness of polymer blends due to that the stiffness and toughness are two mutually exclusive attributes. In this paper, we successfully prepared dynamically vulcanized polypropylene/ethylene-propylene-diene rubber (PP/EPDM) blends with balanced stiffness and toughness through in-situ compatibilization of in-situ formed zinc dimethacrylate (i-ZDMA). The i-ZDMA came from the neutralization of methacrylic acid (MAA) and excess zinc oxide (ZnO) nanoparticles. With i-ZDMA, strong PP/EPDM interface and size reduction of EPDM particles were achieved during peroxide-induced dynamic vulcanization. The average size of EPDM particles was reduced to 260–280 nm, which significantly enlarged PP/EPDM interfacial surfaces. At the same time, the residual excess ZnO nanoparticles migrated into PP phase and served as nucleating agents to increase the crystallinity of PP phase. When the MAA/ZnO reached 2/1.3, the Izod impact strength was increased to 80.1 kJ/m², nearly 1.6 times that of the blend without i-ZDMA and 22 times that of neat PP, while the Young's modulus and yield stress were increased to 373 MPa and 14.8 MPa, respectively, showing an improved rigidity of blend.

1. Introduction

In light of the outstanding chemical resistances, easy processability, suitable thermal stability and low cost, polypropylene (PP) has always been one of the most widely used plastics in civil manufacture, automobile, aviation, electronics, and chemical preservation, etc. [1] However, its fracture toughness, especially the notch impact strength, is not ideal which limits its usage in many applications [2]. Blending with rubbers is an efficient strategy to achieve desirable toughness for PP [3]. There are many factors affect the toughening efficiency of rubbers, for example the phase interfacial interaction, the size and distribution of rubber particles, the content of rubbers, and the distance between rubber particles, etc. [4–7] Among of the above-mentioned factors, a strong plastic/rubber interface is recognized as the essential foundation for achieving considerable toughness [8]. Therefore, ethylene-propylene-diene rubber (EPDM) is considered as one of the most effective tougheners for PP due to that their similar chemical structures facilitate the physical interactions at interface [9].

To achieve a stronger PP/EPDM interface, tremendous attentions

have paid to a magical technical process—dynamic vulcanization, where the in-situ compatibilization can be realized in the presence of peroxides [10]. Naskar and Noordermeer [11] reported the generation of in situ graft links at the PP/EPDM interface during dynamic vulcanization with dicumyl peroxide/triallyl cyanurate. They found that the toughening efficiency of in-situ compatibilizers was better than the premade ones. Lai and Zeng [12] prepared a PP/EPDM/lanthanum trimethacrylate (LaTMA) TPE with good mechanical properties and flame retardancy via peroxide-induced dynamic vulcanization. In-situ grafted LaTMA in the PP and EPDM phase improved their compatibility, which is thought to be the main cause of enhanced mechanical properties. In our previous works [13–15], we also reported the excellent in-situ compatibilization effect of zinc dimethacrylate (ZDMA) on PP/EPDM blends. Peroxides induced the graft polymerizations between ZDMA, PP and EPDM, which generated a powerful interface to improve the mechanical properties of PP/EPDM blends.

Unfortunately, one fact must be faced, that is the presence of rubber inevitably sacrifices the strength and stiffness of plastic [16]. Incorporation of nanofillers into PP phase is a promising approach to

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balance the stiffness and toughness of PP/EPDM blends [17]. Nanofillers such as organoclay [18], carbon nanotubes [19,20] and graphene [21] with large surface areas have a strong interfacial interaction with PP chains, which leads to a dramatic improvement in the thermal stability, mechanical properties of the PP/EPDM blends. The dispersion and location of nanofillers in the nonpolar PP phase are key to affect the property of PP/EPDM blends [22], which depend on a large extent on the mixing process during dynamic vulcanization. Zha and Dang et al. [23] employed polypropylene-graft-maleic anhydride (PP-g-MA) as compatibilizer and prepared PP-g-MA/single-walled carbon nanotubes (SWNTs) masterbatch and PP-g-MA/functionalized graphene sheets (FGS) masterbatch via melt compounding SWNTs and FGS with PP-g-MA, respectively. Then, the masterbatch was blended with PP/EPDM TPV to obtain composites. The SWNTs and FGS thus located in PP phase, and the synergistic effect of SWNTs (0.5 wt%) and graphene (2 wt%) resulted in 82% and 15% increase in tensile modulus and tensile strength respectively for TPV composite. Razavi Aghjeh et al. [24] investigated the localization of multiwall carbon nanotube (MWCNT) on the fracture toughness and deformation mechanisms of un-vulcanized and dynamically vulcanized PP/EPDM blends. They found that the dispersion of MWCNT in the PP matrix is positive to increase the toughness and the maximum tensile loading of blends. Naderi et al. [25,26] prepared dynamically vulcanized PP/EPDM/organoclay composites via various mixing process. Results showed that premixing the nanoclay with EPDM phase resulted in the trapped silicate within EPDM phase after dynamic vulcanization process, while the nanocomposite samples prepared through mixing the preblended PP/clay masterbatch with EPDM show better dispersion of EPDM and clay within the PP phase, which increased the modulus of blends.

Obviously, although tremendous effort has been made in improving the stiffness (adding nanofillers) and toughness (forming a strong phase interface) of PP/EPDM blends, these two effective techniques are relatively independent in dynamic vulcanization. There are rare studies report the in-situ compatibilization technique which can simultaneously improve the stiffness and toughness of dynamically vulcanized PP/EPDM blends. From this point of view, in this paper, we employed ZDMA prepared in-situ from methacrylic acid (MAA) and excess zinc oxide (ZnO) to replace preformed ZDMA as the in-situ compatibilizer for PP/EPDM blends [13,14]. During DCP-induced dynamic vulcanization, in-situ formed ZDMA (i-ZDMA) was also readily graft-polymerized with EPDM and PP chains [13,14], forming a powerful PP/EPDM interface to improve the efficiency of stress delivery between PP continuous phase and crosslinked EPDM particles [15]. At the same time, the EPDM particles were reinforced by the polymerization of i-ZDMA [27] and residual ZnO nanoparticles. The ZnO nanoparticles also contributed to improve the shearing efficiency, which resulted in a further slight size reduction of EPDM particles. As a result, the stronger PP/EPDM interface, as well as the increased plastic/rubber contact area, helped to increase the toughness of blends. More importantly, the stiffness of PP phase was improved by the increased degree of crystallinity through heterogeneous nucleation of migrated residual excess ZnO nanoparticles. Therefore, the stiffness and toughness of dynamically vulcanized PP/EPDM/i-ZDMA blends were simultaneously improved by the synergistic effect of strong PP/EPDM interface, reinforced EPDM particles and strengthened PP phase.

2. Experimental section

2.1. Materials

PP, $T_m = 167^\circ\text{C}$ and $\text{MFI} = 12 \text{ g}/10\text{min}$ (210°C , 2.16 kg), was purchased from CNOOC & Shell Petrochemicals Company Limited; EPDM, with 56% ethylene and 7.5% 5-ethylidene-2-norbornene, density = $0.87 \text{ g}/\text{cm}^3$ and $\text{ML}(1 + 4, 100^\circ\text{C}) = 40$, was purchased from Jilin Chemical Co. (China); Dicumyl peroxide (DCP), purchased from Guangzhou Xinggang Chem factory (China), was purified by anhydrous

alcohol recrystallization before use. ZnO, analytical pure, average size about 50 nm, was purchased from XiLong Chemical Company (Guangzhou, China); MAA, analytical pure, was purchased from Sinopharm Chemical Reagent Co. Ltd (China).

2.2. Preparation of EPDM/i-ZDMA masterbatch

The EPDM/i-ZDMA masterbatch was prepared by two-step method: (i) ZnO nanoparticles were first mixed with EPDM in a two-roller mill for about 5 min. Then, MAA was slowly dropped into EPDM/ZnO masterbatch by an injector and the temperature of rollers was controlled at $45\text{--}50^\circ\text{C}$ to help the neutralization between MAA and ZnO (the time was about 8 min). (ii) The prepared masterbatch was transferred into the mixing chamber of a Haake Rheocord 90 and mixed for another 5 min with the rotor speed of 32 rpm at 60°C . The second step was aimed to further promote the neutralization of MAA and ZnO. The high mixing temperature, the long mixing time and excess ZnO should make ZnO and MAA react to their utmost to form i-ZDMA [28]. More details about the ZDMA prepared in-situ through the reaction of MAA and ZnO during rubber mastication have been reported in Refs. [27–32]. On the basis of the molecular formula of ZDMA that one Zn^{2+} bridged by two α -methylacrylate groups, the theoretical ideal molar ratio of MAA/ZnO was 2/1. Therefore, we fixed the content of i-ZDMA at 20 phr in the EPDM masterbatch by controlling the MAA loading. On this basis, a series of MAA/ZnO molar ratio of 2/1, 2/1.1, 2/1.2 and 2/1.3 were employed to make sure that ZnO nanoparticles were excess.

2.3. Dynamic vulcanization of PP/EPDM/i-ZDMA blends

In this paper, PP/EPDM weight ratio was fixed at 60/40 (w/w). Dynamic vulcanization of PP/EPDM/i-ZDMA blends was occurred in Haake Rheocord 90 at a rotor speed of 50 rpm. The initial temperature was set at 170°C . PP granules were first added into the mixing chamber and shear-melted completely. Then, EPDM/i-ZDMA masterbatch was added. Three minutes later, DCP (1.5 wt% of EPDM) was added to initiate the dynamic vulcanization. After reaching the final stable torque, the dynamically vulcanized blends were removed from the chamber and cooled down to the room temperature. The specimens for mechanical testing were injection molded by an injection molding machine (TTI-160F, Welltec Machinery & Equipment Co. Ltd., China). The nozzle temperature was set at 210°C , the injection pressure was 60 MPa.

2.4. Characterizations

The tensile properties of the dumbbell shaped specimens were carried out in accordance with ISO 527 on a universal testing instrument (Shimadzu AG-1, 10 kN, Japan) at a crosshead speed of 50 mm/min. The notched Izod impact tests were determined according to ISO 180 using a ZWICK5331 (German, Zwick/Roell). Five test specimens were done to calculate the average values for mechanical properties.

Tensile behaviors at 100, 110, 120, 130 and 140°C were carried out in an Instron 5500R machine. The samples were first equilibrated at target temperature for 10 min and then stretched at crosshead speed of 50 mm/min 3 test specimens were done to ensure repeatability.

Dynamic mechanical property was determined on a DMA-Q800 (TA Instruments) under tensile mode from -150 to 150°C at $3^\circ\text{C}/\text{min}$. Frequency was set at 3.33 Hz. The sample size was $15 \times 5.0 \times 1.0 \text{ mm}$. The storage modulus and loss factor were recorded.

SEM observations were done in a Nova NanoSEM 430 (FEI Company, USA). Before observation, the surface of sample was treated by spray-gold.

TEM observations were carried out on a JEM-2100F operating at 200 kV. The specimens were ultramicrotomed into thin sections using a Leica EM UC6 equipment under liquid nitrogen atmosphere.

Crystal morphology of the PP phase were investigated by using an

Axioskop 40 Polarized optical microscopy (POM) (ZEISS, Germany) equipped with a hot stage under crossed polarizers. The samples were first melted at 200 °C and held for 5 min to achieve thermal equilibrium. Then, it was rapidly quenched to 140 °C to induced crystallization.

DSC characterizations were determined using a DSC 240 F1 (NETZSCH, Germany). For non-isothermal crystallization, the samples were heated from room temperature to 200 °C at a rate of 20 °C/min and hold at 200 °C for 5min, then cooled down to 30 °C at a rate of 10 °C/min. Next, the samples were reheated to 200 °C at a heating rate of 20 °C/min. The crystallinity (X_c , %) was determined according to equation (1):

$$D_n = \frac{\sum_i n_i D_i}{\sum_i n_i} \quad (1)$$

where the heat of crystallization (ΔH_c) was determined from the area under the exotherm, the heat of fusion of 100% PP was 209 J/g and ϕ is the mass fraction of PP in the TPV.

3. Results and discussion

3.1. Mechanical properties of dynamically vulcanized PP/EPDM/i-ZDMA blends

As shown in Fig. 1a, although 60/40 (w/w) is a responsible PP/EPDM weight ratio that endows blend with a considerable Izod impact strength of 50.5 kJ/m², the impact strength is further increased to 59.3 kJ/m² after incorporation of i-ZDMA (MAA/ZnO = 2/1). Excitingly, with the increasing MAA/ZnO ratio, the blends exhibit an increased impact strength. When the MAA/ZnO reaches 2/1.3, the Izod impact strength is increased to 80.1 kJ/m², nearly 1.6 times that of the blend without MAA/ZnO and 22 times that of neat PP (3.6 kJ/m², not provided in Fig. 1a). It should be pointed out that all the samples were not impact fractured completely. Therefore, the calculated theoretical impact strengths by considering the un-fractured part further absorbing impact energy are much higher than the experimental values (solid

column), as marked as dash column in Fig. 1a.

Typical stress-strain curves of dynamically vulcanized PP/EPDM/i-ZDMA blends are shown in Fig. 1b, and their Young's modulus, yield stresses and elongations at break are given in Fig. 1c, d and 1e, respectively. As expected, residual excess ZnO nanoparticles reinforced the blends and turned out improved tensile behaviors. Young's modulus, represented the rigidness of material, increases from 320 MPa to 332 MPa when the MAA/ZnO = 2/1 and it further increases to 373 MPa when the MAA/ZnO reaches 2/1.3, showing an improved rigidness (Fig. 1c). The yield stress shows a similar increasing tendency that it increases from 13.2 MPa to 14.8 MPa when the MAA/ZnO = 2/1.3 (Fig. 1d). At least two aspects can be explained the reinforcement of i-ZDMA on blends. First, i-ZDMA is an effective reinforcer for rubbers in the present of peroxides [27]. To address this, the mechanical properties of EPDM/i-ZDMA vulcanizates are summarized in Table S1 (supporting information). The increased strength and hardness of EPDM particle reduced its rigid-softening effect on the PP continuous phase. Second, partial residual ZnO nanoparticles migrated into PP phase and served as heterogeneous nucleating agents, which increased the crystallinity of PP phase and consequently improved the stiffness of blends. This will be discussed later. Interestingly, the extensibility of blends was also increased by increasing the MAA/ZnO ratio (Fig. 1e), which suggested that the interaction between PP and EPDM was improved [33]. A strong interface reduced the risk of rupture at phase interface during stretching [34]. Therefore, the above mechanical performances demonstrate an exciting fact that the impact strength and tensile strength can be simultaneously improved by increasing the MAA/ZnO ratios.

3.2. Size reduction of EPDM particles

In order to observe the morphology of crosslinked EPDM particles, samples were etched by xylene at 100 °C for 20min to remove the PP phase on their surface layers. As shown in Fig. 2a, the EPDM particles in blend without MAA/ZnO are spherical shape with a relative smooth surface, and an average size about 1.7 μm. Incorporation of i-ZDMA significantly changed the morphology of EPDM particles. As shown in Fig. 2b, c, 2d and 2e, the average size of EPDM particles is reduced to

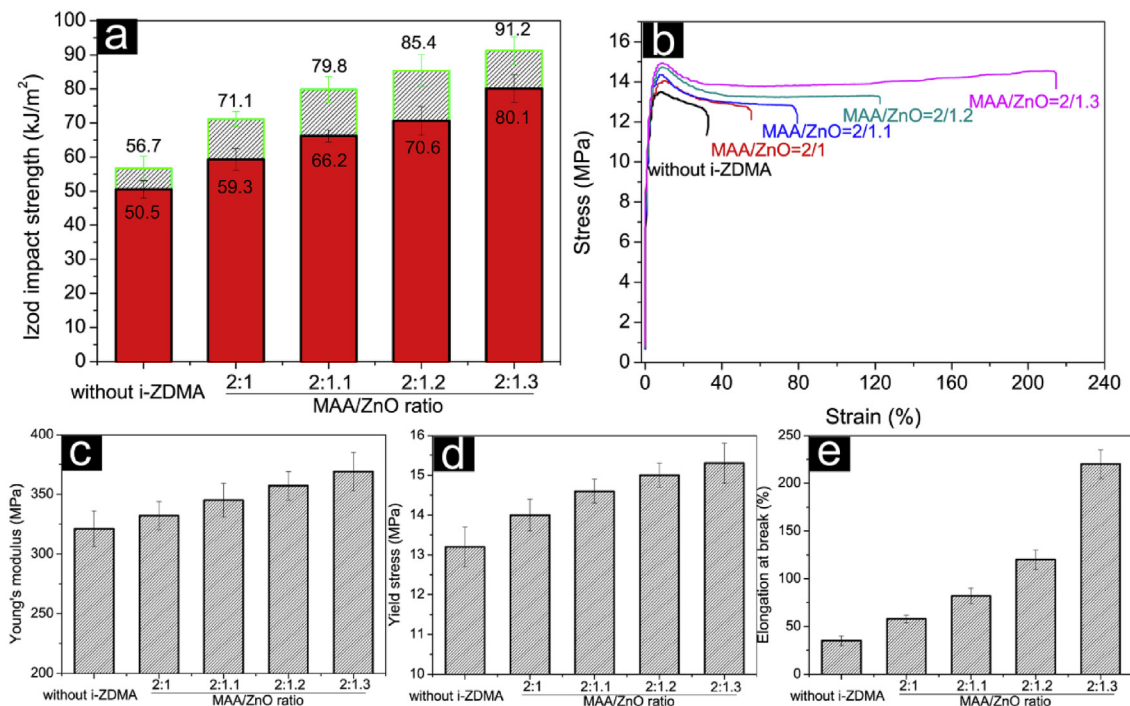


Fig. 1. Mechanical properties of dynamically vulcanized PP/EPDM/i-ZDMA blends: (a) Izod impact strength; (b) typical stress-strain curves; (c) Young's modulus; (d) yield stresses and (e) elongations at break.

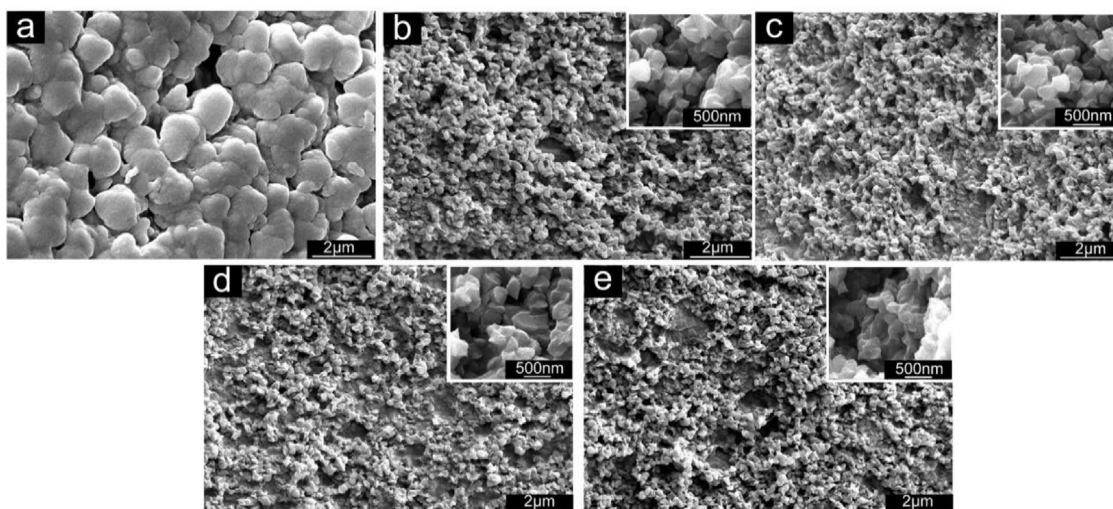


Fig. 2. SEM images of crosslinked EPDM particles in blends: (a) without i-ZDMA; (b) MAA/ZnO = 2/1; (c) MAA/ZnO = 2/1.1; (d) MAA/ZnO = 2/1.2; (e) MAA/ZnO = 2/1.3.

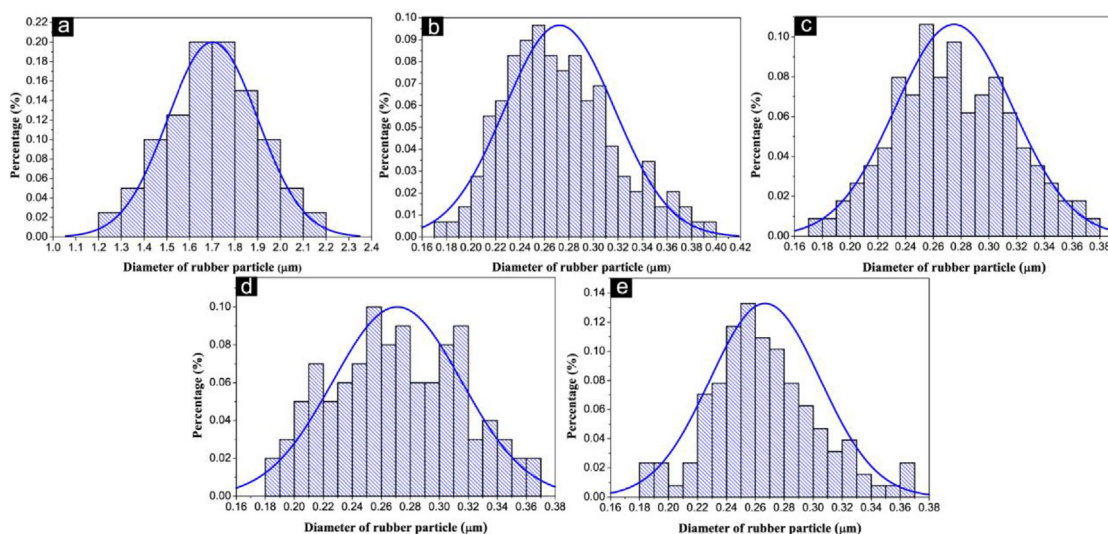


Fig. 3. Size distributions of crosslinked EPDM particles in blends: (a) without i-ZDMA; (b) MAA/ZnO = 2/1; (c) MAA/ZnO = 2/1.1; (d) MAA/ZnO = 2/1.2; (e) MAA/ZnO = 2/1.3.

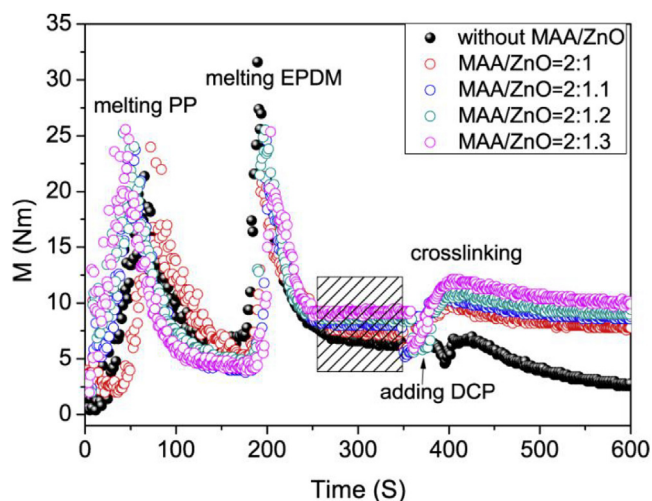


Fig. 4. Torque changes of PP/EPDM/i-ZDMA blends during DCP-induced dynamic vulcanization.

about 260–280 nm, which remarkably enlarges the PP/EPDM interfacial contact area. ZDMA is an effective crosslinking accelerator for rubbers when using peroxides as initiator [32]. The viscosity and elasticity of EPDM phase were quickly increased by i-ZDMA at the initial stage of dynamic vulcanization, which facilitated the rotors shearing EPDM domains into smaller sizes [35]. Then, the promoted crosslinking quickly stabilized the EPDM particles, preventing the second coalescences. However, the shape of EPDM particles became irregular, which might be originated from the mismatching between shearing and accelerated crosslinking rate of EPDM by i-ZDMA [36]. Furthermore, the residual ZnO nanoparticles had not only a positive effect on increasing viscosity and elasticity of EPDM phase but also a barrier effect on coalescence of EPDM particles during melt processing, hence improving the shearing efficiency [37]. This resulted in a further slight size reduction of EPDM particles at higher MAA/ZnO ratios. The size reduction of EPDM particles was definitely critical to improving the toughness of dynamically vulcanized PP/EPDM blends. The corresponding size distributions of EPDM particles are given in Fig. 3.

3.3. In-situ compatibilization of i-ZDMA

In-situ compatibilization of i-ZDMA took place during DCP-induced dynamic vulcanization, which can be analyzed based on the torque changes during preparation of blends. As shown in Fig. 4, the first two main torque peak are attributed to the melting of PP pellets and addition of EPDM/i-ZDMA masterbatch. Herein, it is worth noting that the stable torque at around 5min, representing a full homogenization state of blending components, shows an increase with the increasing MAA/ZnO ratios. This suggested that partial residual excess ZnO nanoparticles successfully migrated into PP phase, which enhanced the melting torques. Before adding DCP, EPDM had been broken-up into small droplets. The scattering and coalescence of EPDM droplets achieved a dynamic equilibrium [35]. After adding DCP, EPDM droplets underwent crosslinking process, leading to a drastic increase in the viscosity and elasticity. This turned out an apparent crosslinking peak in the torque curves at about 6–7min. Apparently, with i-ZDMA, the crosslinking peak had a significant promotion due to that i-ZDMA accelerated the crosslinking rate of EPDM phase [13]. The i-ZDMA promoted the crosslinking of EPDM droplets and transformed them into rubber particles. Besides, in light of the unsaturated double bonds, DCP was readily to initiate the polymerization of i-ZDMA [32], forming macromolecular radicals of polymerized ZDMA (PZDMA). Meanwhile, DCP also induced EPDM and PP macroradicals through hydrogen abstraction [10]. Therefore, the complex in-situ compatibilization reactions between macromolecular radicals of PZDMA, PP and EPDM resulted in a strong PP/EPDM interface. The schematics of preparation of EPDM/PP/i-ZDMA blends and the structure of in situ compatibilized PP/EPDM interface are shown in Fig. 5. The enhanced phase interactions further facilitated the shearing delivery from PP melt to EPDM domains, which was helpful to the size reduction of EPDM particles (Fig. 2) and turned out a higher final torque [38]. The increased friction between EPDM particles and PP phase raised the temperature in the mixing chamber, as shown in Fig. S1. The effect of ZDMA on the promoting the crosslinking of rubbers can be found in Refs. [30–32].

TEM observation further confirmed the in-situ compatibilization in dynamically vulcanized PP/EPDM/i-ZDMA blends. Although it is reported that peroxides would induce in-situ compatibilization of PP and EPDM in some degree, the sharper boundary of EPDM particles in Fig. 6a reveals a relative weak compatibilization effect in the blend

without MAA/ZnO. However, the addition of i-ZDMA greatly deducted the acuity of PP/EPDM interface. It is observed that the boundary of EPDM particles becomes obscured in the blends with MAA/ZnO = 2/1 (Fig. 6b), 2/1.1 (Fig. 6c), 2/1.2 (Fig. 6d) and 2/1.3 (Fig. 6e), suggesting an improved interfacial compatibilization between PP and EPDM. In addition, lots of nanoparticles with size of about 20–50 nm appeared in the blends with i-ZDMA, which were associated with the ZnO nanoparticles and PZDMA nanoparticles [39]. However, the sizes of ZnO and PZDMA are too similar to distinguish them clearly. Nevertheless, based on above experimental result, it can be concluded that the i-ZDMA improved the dynamic vulcanization effect, reducing the acuity of PP/EPDM interface, decreasing particle size and further narrowing down the distribution of particle size (Fig. 3). As a result, the stronger PP/EPDM interface, as well as the increased plastic/rubber contact area, helped to increase the toughness of blends. What should be pointed out is that some small aggregations appear in the sample of blend with MAA/ZnO = 2/1.2 (Fig. 6d). However, the size of those aggregations is about 50–100 nm, which still belongs to nano-scale. This size level usually turns out a positive effect, e.g. reinforcement, on composites rather than deterioration. Therefore, the corresponding mechanical properties was still enhanced (Fig. 1).

The strengthened PP/EPDM interface was verified from the comparison of the cryo-fracture surface morphologies of PP/EPDM/i-ZDMA blend before and after dynamic vulcanization. As shown in Fig. 7a, the un-vulcanized EPDM particles can be easily distinguished from the PP matrix. Furthermore, a lot of EPDM particles were pulled out, leaving hollow holes on the cryo-fracture surface of blend. However, after dynamic vulcanization, no EPDM particles can be found in Fig. 7b, let alone the pulling-out of crosslinked EPDM particles. This comparison strongly suggested that a powerful PP/EPDM interface was achieved in the PP/EPDM/i-ZDMA blend after DCP-induced dynamic vulcanization. To further confirm the in-situ compatibilization effect of i-ZDMA, a SEM image of cryo-fracture surface morphology of dynamically vulcanized PP/EPDM blend without MAA/ZnO was shown as reference in Fig. S2. As clearly seen in Fig. S2, the EPDM phase can be distinguished roughly from PP matrix. Even after DCP-induced dynamic vulcanization, some EPDM particles were still pulled out. This indicated that i-ZDMA was critical to achieving excellent in-situ compatibilized PP/EPDM interface. However, the i-ZDMA compatibilized PP/EPDM interface was weakened at higher temperatures due to that the mass ion pairs in PZDMA were sensitive to temperature [40]. The typical stress-

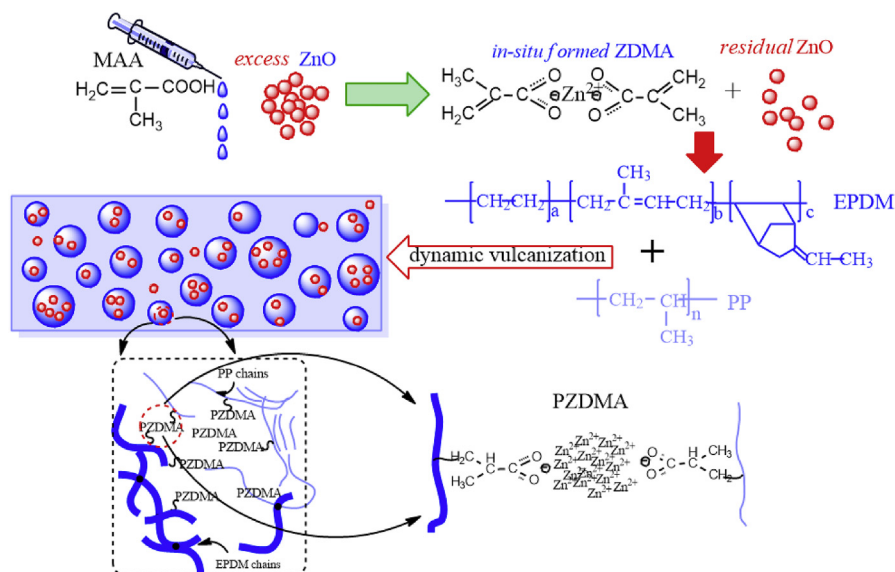


Fig. 5. Schematic illustration of preparation of EPDM/PP/i-ZDMA blends via DCP-induced dynamic vulcanization and structure of in situ compatibilized PP/EPDM interface.

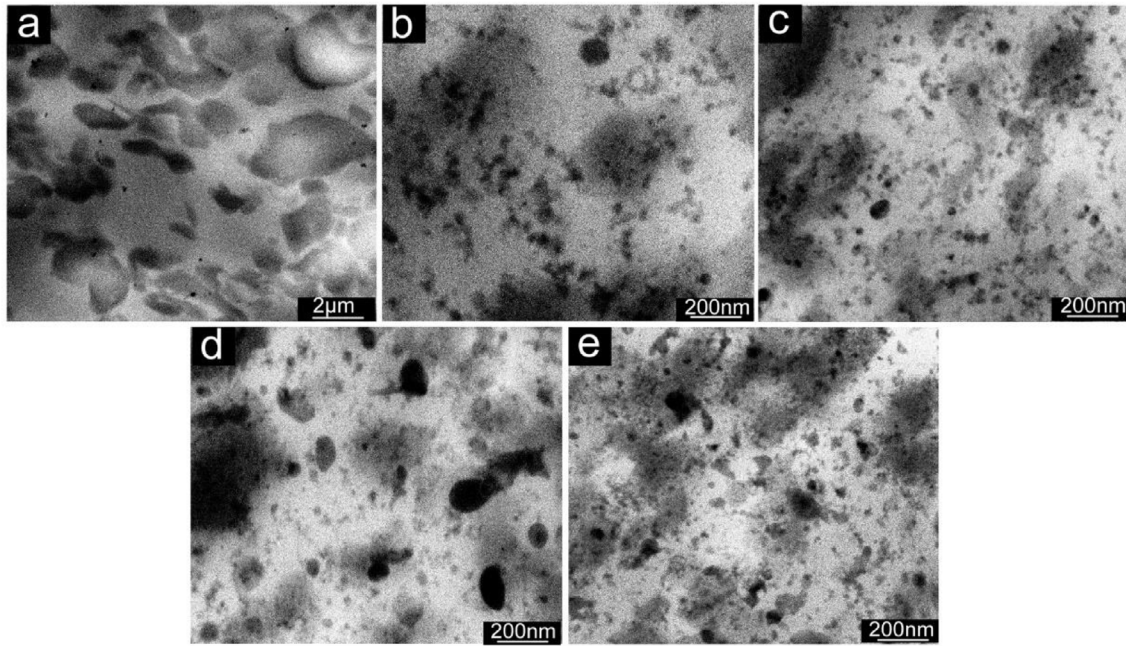


Fig. 6. TEM images of dynamically vulcanized PP/EPDM/i-ZDMA blends: (a) without i-ZDMA; (b) MAA/ZnO = 2/1; (c) MAA/ZnO = 2/1.1; (d) MAA/ZnO = 2/1.2; (e) MAA/ZnO = 2/1.3.

strain curves of dynamically vulcanized PP/EPDM/i-ZDMA blends at high temperatures are shown in Fig. S3. At high temperatures, especially beyond 100 °C, the ionic interactions are deteriorated seriously, which goes against the interactions between PP phase and EPDM particles. This turned out reduced mechanical performance of dynamically vulcanized PP/EPDM/i-ZDMA blends.

3.4. Localization of residual ZnO nanoparticles

In general, at thermodynamic equilibrium, the localization of nanoparticles in a polymer blend tend to require minimizing the interfacial energy [41]. Therefore, in order to support the conclusions made from torque analyses (Fig. 4) regarding the migration of ZnO nanoparticles, the equilibrium position of ZnO was predicted by the evaluation of interfacial tension between possible component pairs. The interfacial tension values can be calculated from the surface tension of components according to Owens-Wendt mean equation (Eq. (2)) [42]. The contact angles and surface tension values of PP, EPDM and ZnO nanoparticles are referenced from literature [39,43,44] and summarized in Table S2.

$$\gamma_{AB} = \gamma_A + \gamma_B - 2(\gamma_A^d \gamma_B^d)^{\frac{1}{2}} - 2(\gamma_A^p \gamma_B^p)^{\frac{1}{2}} \quad (2)$$

where γ_{AB} is the interfacial tension, γ_A and γ_B are the surface tensions of the two materials in contact.

Table 1

The values of interfacial tension (γ_{AB}) for possible component pairs in blend.

Possible pairs	Interfacial tension (γ_{AB}) (mN/m)
PP/EPDM	0.141
PP/ZnO nanoparticles	12.32
EPDM/ZnO nanoparticles	12.10

The calculated values of interfacial tension between possible component pairs are shown in Table 1. The similar interfacial tension values of PP/ZnO pair and EPDM/ZnO pair indicate that the ZnO nanoparticles tend to be located in both PP and EPDM phases at a thermodynamic equilibrium state. This meant that partial residual ZnO nanoparticles had probably migrated from EPDM droplets into PP phase during melt blending [45]. This is also in line with the torque analyses and TEM observations. As a consequence, the migrated ZnO nanoparticles were beneficial to enhance the PP continuous phase. However, it must be pointed out that the above theoretical prediction has a limitation that the i-ZDMA might change the surface tension of EPDM phase. Fortunately, our previous work has confirmed that both PP and EPDM have similar tendency to encapsulate ZDMA during melt blending (γ_{AB} of PP/ZDMA = 22.99 and EPDM/ZDMA = 22.41) [39]. This suggested that the theoretical prediction of the migration of residual ZnO nanoparticles remained reliable. After dynamic

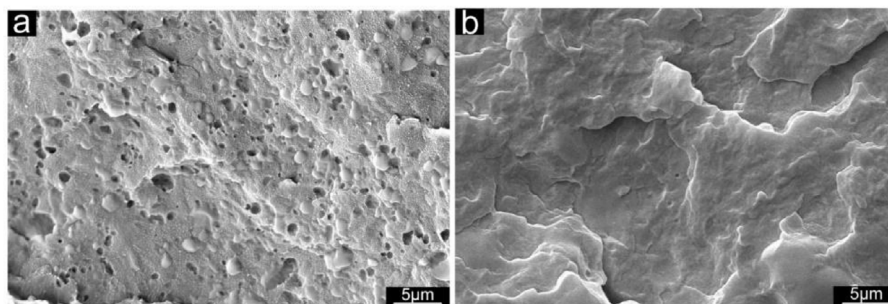


Fig. 7. SEM images of cryo-fracture surface of PP/EPDM/i-ZDMA blend (MAA/ZnO = 2/1.3): (a) un-vulcanized; (b) dynamically vulcanized.

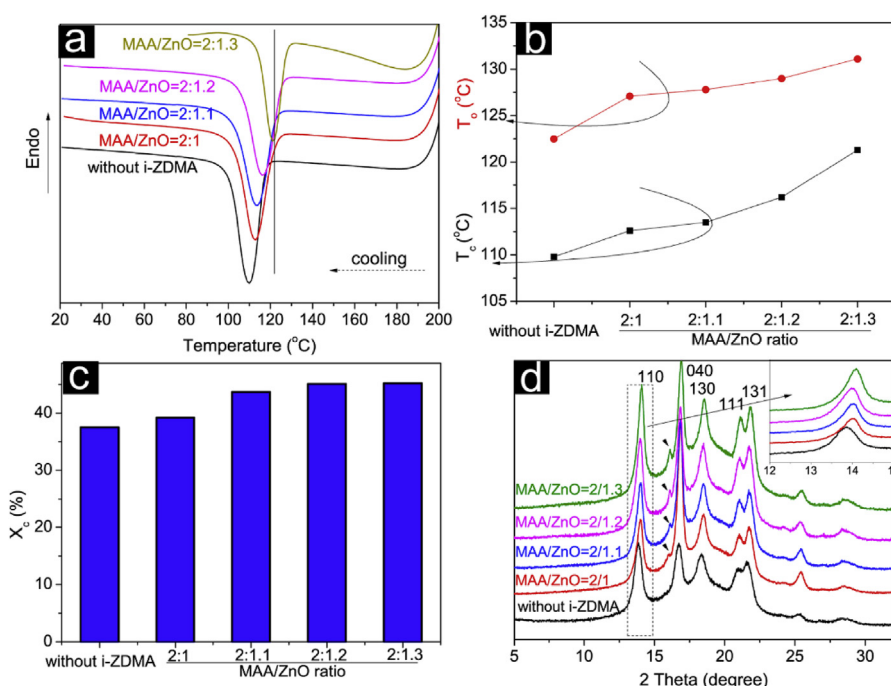


Fig. 8. (a) Cooling DSC curves; (b) onset crystallization temperature and crystallization temperature; (c) crystallinity and (d) WAXD patterns of dynamically vulcanized PP/EPDM/i-ZDMA blends.

vulcanization, the migrated residual ZnO nanoparticles could not back into crosslinked EPDM particle and forever remained in PP phase.

3.5. Crystallization behavior of PP phase in dynamically vulcanized PP/EPDM/i-ZDMA blends

Effects of migrated residual ZnO nanoparticles on the crystallization behavior of dynamically vulcanized PP/EPDM/i-ZDMA blends were investigated by DSC. The cooling DSC curves are shown in Fig. 8a and the heating DSC curves are provided in Fig. S4 (supporting information). Polymerization of i-ZDMA (MAA/ZnO = 2/1) increased the onset crystallization temperature (T_o), crystallization temperature (T_c) and crystallinity (X_c) of PP phase from 109.8 °C, 122.5 °C and 37.5% up to 112.6 °C, 127.1 °C and 39.2%, respectively, due to the heterogeneous nucleation effect of PZDMA nanoparticles. However, the above thermal parameters were further increased with the increasing MAA/ZnO ratio (Fig. 8b and c). For example, the T_o , T_c and X_c of PP phase were further increased to 121.3 °C, 131.1 °C and 45.2% when the MAA/ZnO ratio was 2/1.3. These results suggested that partial residual ZnO nanoparticles were indeed migrated into PP phase and played a role of nucleating agents. The increased X_c should be responsible for the improvement of rigidity of dynamically vulcanized PP/EPDM/i-ZDMA blends. The crystal structures of PP phase were investigated by XRD. As shown in Fig. 8d, WAXD patterns of all the blends show the typical α -crystal diffraction peaks at about 14.0°, 16.9°, 18.5°, 21.0° and 21.9°, corresponding to the (110), (040), (130), (111), and (131) planes of PP phase [46], respectively. However, a new diffraction peak appears at about $2\theta = 16.1^\circ$ in the dynamically vulcanized PP/EPDM/i-ZDMA blends, which corresponds to the (300) planes of β crystal of PP [47]. As seen, the intensity of β crystal diffraction peak increased with the increase of MAA/ZnO ratio, which suggested that the residual ZnO nanoparticles could act as β -crystal nucleating agent for this blend system. Furthermore, the (110) diffraction peak was shifted toward higher 2θ , and the (111) diffraction peak became stronger with the increased residual ZnO nanoparticles, indicating that the spherocrystal was restricted during crystal growth process. To address this, the crystal morphology of the dynamically vulcanized PP/EPDM/i-ZDMA blends

was observed by POM, as shown in Fig. S5 of supporting information. The POM images clearly show that the spherulites were squeezed out during crystal growth, resulting in defects and distortions in the paracrystalline structure, and thus reduced spherulite size in the dynamically vulcanized PP/EPDM/i-ZDMA blends. All these changes in crystal structures of PP phase are favorable to the improvement of toughness of blends.

Based on the experimental data of non-isothermal crystallization, the relative crystallinity (X_t) at crystallization temperature (T) is calculated according to the follow equation [48]:

$$X_t = \int_{T_0}^T \left(\frac{dH}{dT} \right) dT / \int_{T_0}^{T_\infty} \left(\frac{dH}{dT} \right) dT \quad (3)$$

where T_0 and T_∞ represent the temperature at the initial and end of crystallization, respectively, and dH/dT is the heat flow at temperature T . The X_t of blends V.S. temperature is shown in Fig. 9a. The crystallization temperature shifts toward higher temperature and the crystallization temperature zone becomes narrow with increasing the MAA/

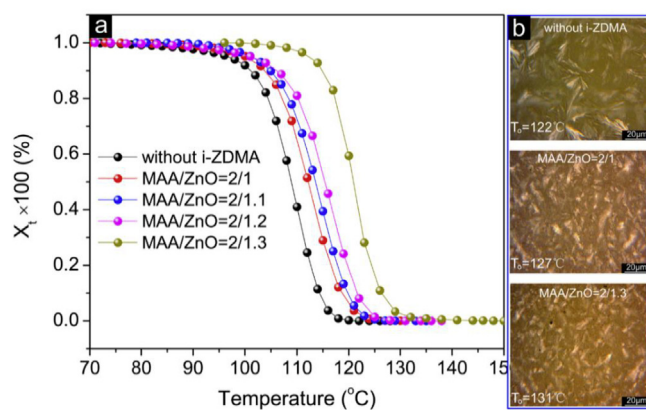


Fig. 9. (a) X_t of dynamically vulcanized PP/EPDM/i-ZDMA blends V.S. temperature; (b) POM images of blends without i-ZDMA, MAA/ZnO = 2/1 and MAA/ZnO = 2/1.3.

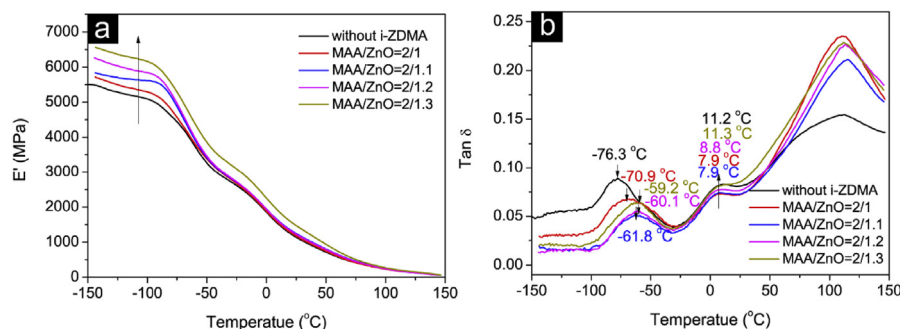


Fig. 10. Temperature dependence of (a) storage modulus (E') and (b) loss factor ($\tan \delta$) of dynamically vulcanized PP/EPDM/i-ZDMA blends.

ZnO ratio, suggesting that residual ZnO nanoparticles promoted the crystallization rate in the non-isothermal crystallization. Because of the limitation of POM instrument, we only obtained the POM images at a constant temperature. The POM images of blends at their T_o for 1min are shown in Fig. 9b. This further confirmed that the increased ZnO nanoparticles in PP phase served as nucleating agents.

3.6. Dynamic mechanical properties of dynamically vulcanized PP/EPDM/i-ZDMA blends

The temperature dependence of E' of dynamically vulcanized PP/EPDM/i-ZDMA blends is shown in Fig. 10a. It is clearly seen that the E' , which provides insight into the stiffness of blends, is enhanced by i-ZDMA in the whole temperature window, especially in the glass region. For example, the blend without i-ZDMA shows an E' of about 5100 MPa at -100°C ; however, the blend with MAA/ZnO = 2/1 possesses an E' of 5323 MPa. Even more strikingly, when the MAA/ZnO reaches 2/1.3, the E' dramatically increases to about 6180 MPa. The considerably increased E' could be explained by the reinforced EPDM particles with higher hardness and the strengthened PP phase with higher stiffness at their glass states, which agrees well with the enhanced mechanical performance in Fig. 1. It is interesting to discuss the shift of $\tan \delta$ peak temperatures of EPDM and PP phases in Fig. 10b. For the EPDM phase, the shift of $\tan \delta$ peak temperature, which is associated to the glass-to-rubber transition temperature (T_g), can be used to confirm the promoted crosslink density of rubber particles [49,50]. Compared with the blend without i-ZDMA, the blend with MAA/ZnO = 2/1 shows an increase of $\tan \delta$ peak temperature from -76.3 to -70.9°C due to the reinforcement of polymerization of i-ZDMA. With increasing the MAA/ZnO ratio, the $\tan \delta$ peak temperature is further shifted to -61.8°C for MAA/ZnO = 2/1.1, -60.1°C for MAA/ZnO = 2/1.2 and -59.2°C for MAA/ZnO = 2/1.3, respectively. This strongly confirmed that the residual ZnO nanoparticles had an additional restricting effect on the chain mobility of EPDM. As for the PP phase, the $\tan \delta$ peak temperature decreases from 11.2 to 7.9°C due to

the in-situ compatibilization of i-ZDMA; however, it is shifted to 11.3°C when the MAA/ZnO increases to 2/1.3. This could be explained as the heterogeneous nucleation effect of migrated ZnO nanoparticles that increased the crystallinity of PP as discussed before.

3.7. A simple discussion on the toughening and extensibility of dynamically vulcanized PP/EPDM/i-ZDMA blends

According to the classical report of Wu [51], a critical interparticle distance (τ_c) played an important role in toughening a polymer/rubber blend.

$$\tau = d_c \left[\left(\frac{\pi}{6\phi_r} \right)^{\frac{1}{3}} - 1 \right] \quad (4)$$

where ϕ_r is the volume fraction of rubber component, τ is the average surface-to-surface interparticle distance which is depended on the ϕ_r . In this PP/EPDM/i-ZDMA blend system, i-ZDMA graft-polymerized onto EPDM molecules during DCP-induced dynamic vulcanization, which increased the effective volume of EPDM. In addition, the partial residual ZnO nanoparticles remained in EPDM particles also added additional contributions to this effect. Furthermore, size reduction and consequent dramatically enlarged surface area of EPDM particles effectively decreased the τ , which facilitated the continuum percolation of stress volume around EPDM particles and the propagation of stress yielding in the PP continuous phase [52].

Beside the above mentioned τ , the strengthened PP/EPDM interface must be considered regarding the toughness and the extensibility of dynamically vulcanized PP/EPDM/i-ZDMA blends. Fig. 11 a and b show the impact-fractured surfaces of blends with MAA/ZnO = 2/1 and 2/1.3, respectively. Many tentacle-like structures along the impact orientation are clearly observed on the impact-fractured surfaces of both samples, which are associated with the deformed EPDM particles combined with plastic deformation of PP phase since no rubber particles are found on the impact-fractured surfaces [15]. Note that these

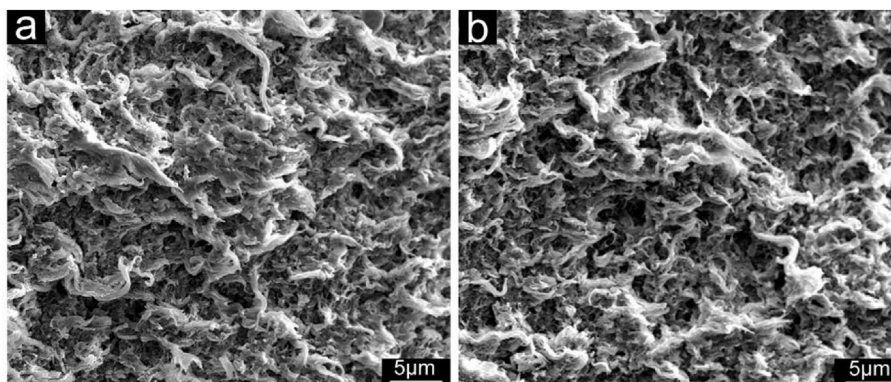


Fig. 11. SEM images of impact-fractured surfaces of blend with (a) MAA/ZnO = 2/1 and (b) 2/1.3.

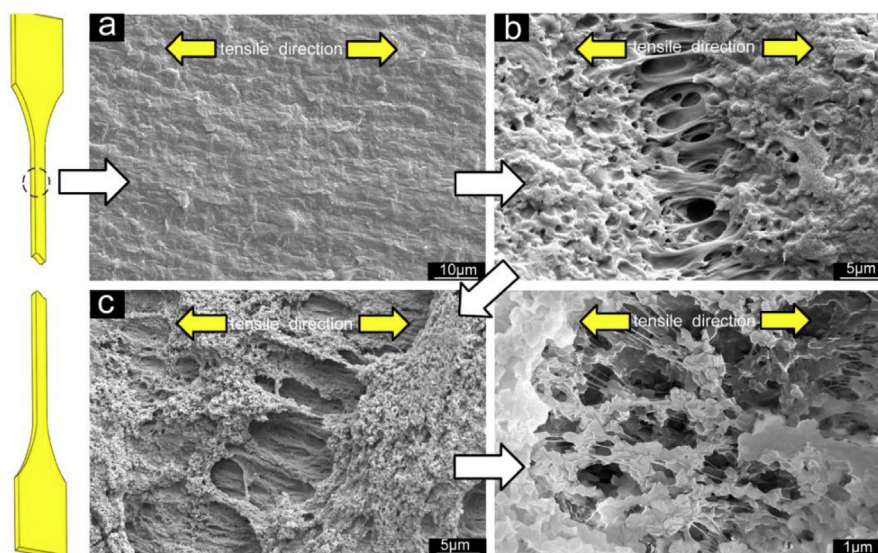


Fig. 12. SEM images of (a) longitudinal stretching surface of the blend with MAA/ZnO = 2/1.3; the longitudinal stretching surface was etched by xylene at 60 °C for (b) 2min, (c) 5min and (d) 8min.

tentacle-like structures were firmly adhered on the fractured surface, showing an excellent interfacial adhesion. This suggested that the powerful PP/EPDM interface delivered impact stress efficiently from PP continuous phase to EPDM particles when suffering impact. The plastic deformation of PP, through the i-ZDMA in-situ compatibilized PP/EPDM interface, forced the deformation of EPDM particles. During this process, considerable impact energies could be absorbed by EPDM particles and the plastic deformation of PP around the interface. As a result, the formation of this specific morphology stopped the further development of the crazes, enabling the blends to sustain higher impact stress.

To better understand the function of compatibilized PP/EPDM interface in toughening blends, the morphology of longitudinal stretching surface of the blend with MAA/ZnO = 2/1.3 was observed by SEM. The SEM image of the longitudinal stretching surface of the blend without i-ZDMA is proved in Fig. S6 (supporting information). It is clearly seen that many folds appear on the blend without i-ZDMA due to the separated phases slipping at PP/EPDM interfaces. However, as shown in Fig. 12a, no cracks appear on the longitudinal stretching surface of the blend with MAA/ZnO = 2/1.3, showing the excellent interfacial adhesion between PP and EPDM phases. Some evidences of orientation of the EPDM phase can be found in Fig. 12a. After gradually removing the PP phase, the elongated EPDM particles are exposed and recoiled as particles (Fig. 12b, c and d). This suggested that the EPDM particles underwent elongating in the PP continuous phase under the stress of strong PP/EPDM interface.

To address this, the TEM observations were done on the middle part of elongated blends which was ultramicrotomed into thin pieces along

the longitudinal deforming direction. As shown in Fig. 13, EPDM particles are highly elongated, at the same time, no cracks are observed at the PP/EPDM interface, strongly confirming a powerful compatibilized PP/EPDM interface. When stretching, the thickness and width of parallel area of the middle part of dumbbell-shaped samples were continuously decreased. The orientation of PP phase forced the EPDM particle to be elongated and oriented along the stretching direction via the enhanced PP/EPDM interface. This is also applicable to the impact deformation. Therefore, the rubber particles, nuzzling each other, with strong interaction with PP via the strong interface, can be easily deform by force, resulting high impact strength.

4. Conclusion

In this paper, we employed i-ZDMA as in-situ compatibilizer for PP/EPDM blends. With i-ZDMA, a powerful PP/EPDM interface was formed after DCP-induced dynamic vulcanization. The ZnO nanoparticles also contributed to improve the shearing efficiency, which resulted in a further slight size reduction of EPDM particles. The size reduction of EPDM particles significantly enlarged interfacial surfaces. The hardness and strength of EPDM particles were improved by the polymerization of i-ZDMA and the trapped residual excess ZnO nanoparticles while the stiffness of PP phase was improved by the increased degree of crystallinity through heterogeneous nucleation of migrated ZnO nanoparticles. The powerful PP/EPDM interface could deliver stress efficiently from PP continuous phase to EPDM particles under loading. As a result, the stiffness and toughness of dynamically vulcanized PP/EPDM/i-ZDMA blends were simultaneously improved by the synergistic effect of strong

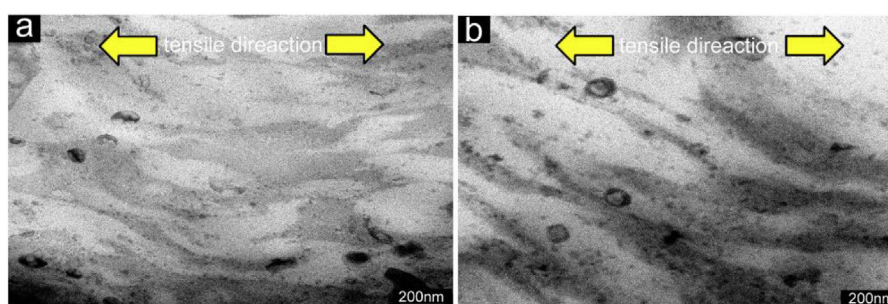


Fig. 13. TEM images of the middle part of elongated blends with (a) MAA/ZnO = 2/1 and (b) MAA/ZnO = 2/1.3.

PP/EPDM interface, reinforced EPDM particles and strengthened PP phase. When the MAA/ZnO = 2/1.3, the Izod impact strength, Young's modulus and yield stress could achieve 80.1 kJ/m², 373 MPa and 14.8 MPa, respectively.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compositesb.2018.10.014>.

References

- Wang ZJ, Kwon DJ, Gu G-Y, Kim HS, Kim DS, Lee CS, Lawrence DeVries K, Park JM. Mechanical and interfacial evaluation of CNT/polypropylene composites and monitoring of damage using electrical resistance measurements. *Compos Sci Technol* 2013;81:69–75.
- Yu BW, Geng CZ, Zhou M, Bai HW, Fu Q, He BB. Impact toughness of polypropylene/glass fiber composites: interplay between intrinsic toughening and extrinsic toughening. *Composites, Part B* 2016;92:413–9.
- Zebarjad SM, Lazzeri A, Bagheri R, Seyed Reihani SM, Frounchi M. Fracture mechanism under dynamic loading of elastomer-modified polypropylene. *Mater Lett* 2003;57(18):2733–41.
- Behazin E, Misra M, Mohanty AK. Compatibilization of toughened polypropylene/biocarbon biocomposites: a full factorial design optimization of mechanical properties. *Polym Test* 2017;61:364–72.
- Fasihi M, Mansouri H. Effect of rubber interparticle distance distribution on toughening behavior of thermoplastic polyolefin elastomer toughened polypropylene. *J Appl Polym Sci* 2016;133:44068.
- Shangguan YG, Chen F, Yang J, Jia EW, Zheng Q. A new approach to fabricate polypropylene alloy with excellent low-temperature toughness and balanced toughness-rigidity through unmatched thermal expansion coefficients between components. *Polymer* 2017;112:318–24.
- Chen F, Qiu B, Wang B, Shangguan Y, Zheng Q. Balanced toughening and strengthening of ethylene-propylene rubber toughened isotactic polypropylene using a poly(styrene-*b*-ethylene-propylene) diblock copolymer. *RSC Adv* 2015;5:20831–7.
- Lim JW, Hassan A, Rahmat AR, Wahit MU. Rubber-toughened polypropylene nanocomposite: effect of polyethylene octene copolymer on mechanical properties and phase morphology. *J Appl Polym Sci* 2006;99:3441–50.
- Ishikawa M, Sugimoto M, Inoune T. Mechanism of toughening for polypropylene blended with ethylene-propylene-diene rubber following selective crosslinking. *J Appl Polym Sci* 1996;62(10):1495–502.
- Saffari A, Sheikh A. Peroxide dynamic crosslinking in impact modification of polypropylene with polybutadiene. *Polym Test* 2017;57:260–9.
- Naskar K, Noordermeer JWM. Influence of premade and in situ compatibilizers in polypropylene/ethylene-propylene-diene terpolymer thermoplastic elastomeric olefins and thermoplastic vulcanizates. *J Appl Polym Sci* 2006;100:3877–88.
- Zhou Q, Yan L, Lai XJ, Li HQ, Zeng XR. The effect of lanthanum trimethacrylate on the mechanical properties and flame retardancy of dynamically vulcanized PP/EPDM thermoplastic vulcanizates. *J Elastomers Plastics* 2018;50:339–53.
- Chen YK, Xu CH, Liang XQ, Cao LM. In situ reactive compatibilization of polypropylene/ethylene-propylene-diene monomer thermoplastic vulcanizate by zinc dimethacrylate via peroxide-induced dynamic vulcanization. *J Phys Chem B* 2013;117:10619–28.
- Chen YK, Xu CH, Cao LM, Wang YP, Fang LM. Morphology study of peroxide-induced dynamically vulcanized polypropylene/ethylene-propylene-diene monomer/zinc dimethacrylate blends during tensile deformation. *J Phys Chem B* 2013;117:7819–25.
- Xu CH, Lin BF, Liang XQ, Chen YK. Zinc dimethacrylate induced in situ interfacial compatibilization turns EPDM/PP TPVs into a shape memory material. *Ind Eng Chem Res* 2016;55:4539–48.
- Liu YT, Cao LM, Yuan DS, Chen YK. Design of super-tough PLA/NR/SiO₂ TPVs with balanced stiffness and toughness based on reinforced rubber and interfacial compatibilization. *Compos Sci Technol* 2018;165:231–9.
- Yang H, Zhang X, Qu C, Li B, Zhang L, Zhang Q, Fu Q. Largely improved toughness of PP/EPDM blends by adding nano-SiO₂ particles. *Polymer* 2007;48:860–9.
- Ghoreishy MHR, Naderi G, Pahlavan M. An investigation into the thermal transport properties of PP/EPDM/clay nanocomposites using a new combined experimental/numerical method. *Plast. Rubber Compos.* 2016;45:229–37.
- Khodabandelou M, Razavi Aghjeh MK, Mehrabi Mazidi M. Fracture toughness and failure mechanisms in un-vulcanized and dynamically vulcanized PP/EPDM/MWCNT blend-nanocomposites. *RSC Adv* 2015;5:70818–31.
- Khodabandelou M, Razavi Aghjeh MK. Impact behavior of CNT filled PP/EPDM blends; effect of dynamic vulcanization and PP-g-MA compatibilizer. *Polym Bull* 2016;73(6):1607–26.
- Zhang YH, Rhee KY, Park SJ. Nanodiamond nanocluster-decorated graphene oxide/epoxy nanocomposites with enhanced mechanical behavior and thermal stability. *Composites Part B* 2017;114:111–20.
- Thankappan Nair S, Poornima Vijayan P, Xavier P, Bose S, George SC, Thomas S. Selective localisation of multi walled carbon nanotubes in polypropylene/natural rubber blends to reduce the percolation threshold. *Compos Sci Technol* 2015;116:9–17.
- Li CQ, Zha JW, Li ZJ, Zhang DL, Wang SJ, Dang ZM. Towards balanced mechanical and electrical properties of thermoplastic vulcanizates composites via unique synergistic effects of single-walled carbon nanotubes and graphene. *Compos Sci Technol* 2018;157:134–43.
- Khodabandelou M, Aghjeh MKR, Khonakdar HA, Mazidi MM. Effect of localization of carbon nanotubes on fracture behavior of un-vulcanized and dynamically vulcanized PP/EPDM/MWCNT blend-nanocomposites. *Compos Sci Technol* 2017;149:134–48.
- Naderi G, Khosrokhavar R, Shokooi S, Bakhshandeh GR, Ghoreishy MHR. Dynamically vulcanized polypropylene/ethylene-propylene diene monomer/organoclay nanocomposites: effect of mixing sequence on structural, rheological, and mechanical properties. *J Vinyl Addit Technol* 2016;22:320–5.
- Nakhaei MR, Mostafapour A, Naderi G. Optimization of mechanical properties of PP/EPDM/clay nanocomposite fabricated by friction stir processing with response surface methodology and neural networks. *Polym Compos* 2017;38:421–32.
- Peng ZL, Liang X, Zhang YX, Zhang Y. Reinforcement of EPDM by in situ prepared zinc dimethacrylate. *J Appl Polym Sci* 2002;84:1339–45.
- Lu YL, Liu L, Tian M, Geng HP, Zhang LQ. Study on mechanical properties of elastomers reinforced by zinc dimethacrylate. *Eur Polym J* 2005;41:589–98.
- Liu C, Huang JR, Yuan DS, Chen YK. Design of high strength XSBR/Fe₃O₄/ZDMA shape memory composite with dual responses. *Ind Eng Chem Res* 2018. <https://doi.org/10.1021/acs.iecr.8b03377>.
- Yuan XH, Peng ZL, Zhang Y, Zhang YX. In situ preparation of zinc salts of unsaturated carboxylic acids to reinforce NBR. *J Appl Polym Sci* 2000;77:2740–8.
- Lu YL, Liu L, Shen DY, Yang C, Zhang LQ. Infrared study on in situ polymerization of zinc dimethacrylate in poly(α -octylene-co-ethylene) elastomer. *Polym Int* 2004;53:802–8.
- Nie YJ, Huang GS, Qu LL, Zhang P, Weng GS, Wu JR. Cure kinetics and morphology of natural rubber reinforced by the in situ polymerization of zinc dimethacrylate. *J Appl Polym Sci* 2010;115:99–106.
- Xu CH, Wu WC, Zheng ZJ, Wang ZW, Nie JD. Design of shape-memory materials based on sea-island structured EPDM/PP TPVs via in-situ compatibilization of methacrylic acid and excess zinc oxide nanoparticles. *Compos Sci Technol* 2018;167:431–9.
- Wang Y, Zhang Q, Na B, Du RN, Fu Q, Shen KZ. Dependence of impact strength on the fracture propagation direction in dynamic packing injection molded PP/EPDM blends. *Polymer* 2003;44:4261–71.
- Li SQ, Lv YF, Sheng J, Tian HC, Ning N.Y, Zhang LQ, Wu HG, Tian M. Morphology development of POE/PP thermoplastic vulcanizates (TPVs) during dynamic vulcanization. *Eur Polym J* 2017;93:590–601.
- Zhao YS, Liu ZW, Su B, Chen F, Fu Q, Ning NY, Tian M. Property enhancement of PP-EPDM thermoplastic vulcanizates via shear-induced break-up of nano-rubber aggregates and molecular orientation of the matrix. *Polymer* 2015;63:170–8.
- Mirzadeh A, Laffeur PG, Kamal MR, Dubois C. The effects of nanoclay dispersion levels and processing parameters on the dynamic vulcanization of TPV nanocomposites based on PP/EPDM prepared by reactive extrusion. *Polym Eng Sci* 2012;52:1099–110.
- Zitzumbo-Guzman R, Avalos-Belmontes F, Ramos-De Valle L, Ortiz-Cisneros J, Alonso-Romero S, Estrada-Monje A. Rheological behavior in the transient state of PP/EPDM blends with carbon nanofillers. *Rheol Acta* 2018;57:397–404.
- Chen YK, Xu CH, Cao LM, Cao XD. Highly toughened polypropylene/ethylene-propylene-diene monomer/zinc dimethacrylate ternary blends prepared via peroxide-induced dynamic vulcanization. *Mater Chem Phys* 2013;138:63–71.
- Gonzalez-Jimenez A, Malmierca MA, Bernal-Ortega P, Posadas P, Perez-Aparicio R, Marcos-Fernandez A, Mather PT. The shape-memory effect in ionic elastomers: fixation through ionic interactions. *Soft Matter* 2017;13:2983–94.
- Wu S. *Polymer interface and adhesion*. New York: Marcel Dekker; 1982.
- Owens DK, Wendt RC. Estimation of the surface free energy of polymers. *J Appl Polym Sci* 1969;13:1741–7.
- Helal E, Pottier C, David E, Fr  chette M, Demarquette NR. Polyethylene/thermoplastic elastomer/Zinc Oxide nanocomposites for high voltage insulation applications: dielectric, mechanical and rheological behavior. *Eur Polym J* 2018;100:258–69.
- Torchinsky I, Rosenman G. Wettability modification of nanomaterials by low-energy electron flux. *Nanoscale Res. Lett.* 2009;4:1209–17.
- Uthairan N, Jarnthong M, Peng Z, Junhasavasdikul B, Nakason C, Thitithammawong A. Micro-scale morphologies of EPDM/EOC/PP ternary blends: relating experiments to predictive theories of dispersion in melt mixing. *Mater Des* 2016;100:19–29.
- Banerjee SS, Bhowmick AK. An effective strategy to develop nanostructured morphology and enhanced physico-mechanical properties of PP/EPDM thermoplastic elastomers. *J Mater Sci* 2016;51:6722–34.
- Tang XG, Bao RY, Yang W, Xie BH, Yang MB, Hou M. Effect of β -phase on the fracture behavior of dynamically vulcanized PP/EPDM blends studied by the essential work of fracture approach. *Eur Polym J* 2009;45(5):1448–53.
- Zeng JB, Srinivasan M, Li SL, Narayan R, Wang YZ. Non-isothermal and isothermal

- cold crystallization behaviors of biodegradable poly(p-dioxanone). *Ind Eng Chem Res* 2011;50:4471–7.
- [49] Cao LM, Yuan DS, Fu XF, Chen YK. Green method to reinforce natural rubber with tunicate cellulose nanocrystals via one-pot reaction. *Cellulose* 2018;25:4551–63.
- [50] Cao LM, Huang JR, Chen YK. Dual cross-linked epoxidized natural rubber reinforced by tunicate cellulose nanocrystals with improved strength and extensibility. *ACS Sustainable Chem Eng* 2018. <https://doi.org/10.1021/acssuschemeng.8b03331>.
- [51] Wu S. A generalized criterion for rubber toughening: the critical matrix ligament thickness. *J Appl Polym Sci* 1988;35:549–61.
- [52] Yang CJ, Huang T, Yang JH, Zhang N, Wang Y, Zhou ZW. Carbon nanotubes induced brittle-ductile transition behavior of the polypropylene/ethylene-propylene-diene terpolymer blends. *Compos Sci Technol* 2017;139:109–16.